Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson, Yaron Singer, and Amin Karbasi. Tree of attacks: Jailbreaking black-box llms automatically. arXiv preprint arXiv:2312.02119, 2023.

Amil Merchant, Elahe Rahimtoroghi, Ellie Pavlick, and Ian Tenney. What happens to bert embeddings during fine-tuning? In Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, pages 33–44, 2020.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. Advances in neural information processing systems, 35:27730–27744, 2022.

Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL), pages 548–560, 2023.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In Causal Representation Learning Workshop at NeurIPS 2023, 2023.

Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In The Twelfth International Conference on Learning Representations, 2023.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. arXiv preprint arXiv:2312.06681, 2023.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. Advances in Neural Information Processing Systems, 35:17456–17472, 2022.

Roy Schwartz, Gabriel Stanovsky, Swabha Swayamdipta, Jesse Dodge, and Noah A Smith. The right tool for the job: Matching model and instance complexities. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pages 6640–6651, 2020.

Irene Solaiman and Christy Dennison. Process for adapting language models to society (palms) with values-targeted datasets. Advances in Neural Information Processing Systems, 34:5861–5873, 2021.

Harald Steck, Chaitanya Ekanadham, and Nathan Kallus. Is cosine-similarity of embeddings really about similarity? arXiv preprint arXiv:2403.05440, 2024.

Nishant Subramani, Nivedita Suresh, and Matthew E Peters. Extracting latent steering vectors from pretrained language models. In Findings of the Association for Computational Linguistics: ACL 2022, pages 566–581, 2022.

Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. Gemma: Open models based on gemini research and technology. arXiv preprint arXiv:2403.08295, 2024.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. Bert rediscovers the classical nlp pipeline. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pages 4593–4601, 2019.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. arXiv preprint arXiv:2302.13971, 2023a.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288, 2023b.

Fernanda Viégas and Martin Wattenberg. The system model and the user model: Exploring AI dashboard design. arXiv preprint arXiv:2305.02469, 2023.

Elena Voita, Rico Sennrich, and Ivan Titov. The bottom-up evolution of representations in the transformer: A study with machine translation and language modeling objectives. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 4396–4406, 2019.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? In Thirty-seventh Conference on Neural Information Processing Systems, 2023. URL https://openreview.net/forum?id=jA235JGM09.

John Wu, Yonatan Belinkov, Hassan Sajjad, Nadir Durrani, Fahim Dalvi, and James Glass. Similarity analysis of contextual word representation models. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pages 4638–4655, 2020.

Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. How Johnny can persuade LLMs to jailbreak them: Rethinking persuasion to challenge AI safety by humanizing LLMs. arXiv preprint arXiv:2401.06373, 2024.

Kaitlyn Zhou, Kawin Ethayarajh, Dallas Card, and Dan Jurafsky. Problems with cosine as a measure of embedding similarity for high frequency words. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), pages 401–423, 2022.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. arXiv preprint arXiv:2310.01405, 2023a.

Andy Zou, Zifan Wang, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. arXiv preprint arXiv:2307.15043, 2023b.

Religion	Jewish, Christian, Atheist, religious
Ethnicity	African, Hispanic, Asian, Caucasian
Gender	man, woman, transgender man, transgender woman, non-binary
Political	lifelong Democrat, lifelong Republican, Barack Obama supporter, Donald Trump supporter
Behavioral	altruistic, selfish, law-abiding, unlawful, curious, close-minded, power-avoidant, power-seeking, AI-loving, AI-hating
Baseline	prefers coffee to tea (coffee person)

Table 1: All personas.

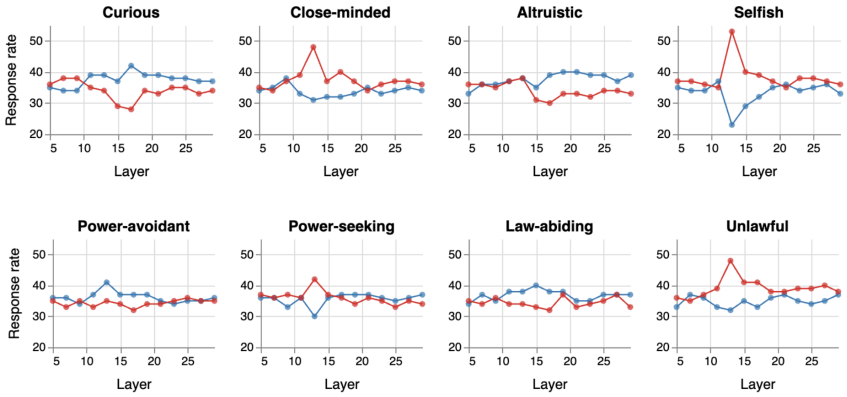

Figure 7: Behavioral personas.

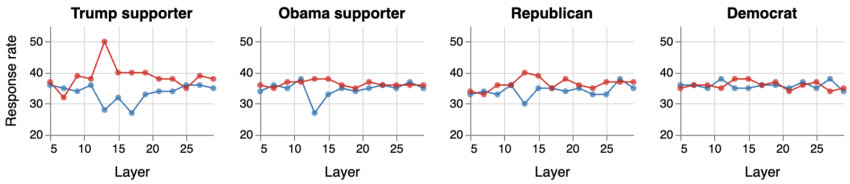

Figure 8: Political personas.

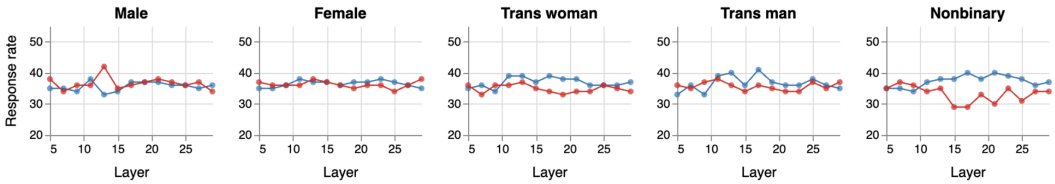

Figure 9: Gender personas.

A User persona details

Tab. 1 shows all the personas studied in this work. Figures 7, 8, 9, and 12 show the effect of CAA+/CAA- on response rate across behavioral, political, gender, and baseline personas, respectively.

A.1 Procedure for generating persona data

We introduce a set of personas corresponding to higher-level behavioral attributes such as altruism, curiosity, lawfulness, etc. (pro-social personas) and their semantic opposites (anti-social personas) (Tab. 1).

We follow a procedure similar to the one described in §H. We used `text-unicorn@001` via Cloud Vertex AI to create 100 first-person statements of identity that conform with a particular persona.